

Classical existence question

Does a given manifold admit a Riemannian metric of positive scalar curvature (pscm)?

Let M be a closed connected spin manifold of dimension m with fundamental group π .

$\hat{A}$ -genus: $\hat{A}(M) = \text{ind } \mathcal{D} \in \mathbb{Z}$.

► spinor bundle of $M \rightsquigarrow$ spin Dirac operator $\mathcal{D}$.

Rosenberg index: $\alpha(M) := \text{ind } \mathcal{D}_{\mathcal{L}} \in KO_m(C^*\pi)$ (higher index).

► $SM \otimes \mathcal{L}(M) \rightsquigarrow$ twisted spin Dirac operator $\mathcal{D}_{\mathcal{L}}$.

► SM : $\mathbb{C}l_m$ -linear spinor bundle

► $\mathcal{L}(M) := \tilde{M} \times_{\pi} C^*\pi$ Mishchenko-Fomenko bundle.

► $C^*\pi$: maximal group C^* -algebra of π .

Fact. The Rosenberg index is the most general known index-theoretical obstruction to the existence of a pscm on M .

$$\begin{array}{c} \hat{A}(M) \neq 0 \implies \alpha(M) \neq 0 \implies \text{no pscm on } M \\ \nearrow \quad \quad \quad \uparrow \\ \dots \quad \quad \quad M \text{ enlargeable} \end{array}$$

Examples. ► The $\hat{A}$ -genus of the n -torus is zero but the Rosenberg index does not vanish.

► There exist exotic spheres Σ^k of non-zero Rosenberg index.

► Aspherical spin manifolds whose fundamental group satisfies the Novikov conjecture have non-vanishing Rosenberg index.

General principle classical index $\rightsquigarrow$ replace by higher index

Classically: A non-vanishing $\hat{A}$ -genus gives rise to a non-vanishing $\mathcal{D}$ -harmonic spinor u . In extreme geometric situations u is parallel, hence

$$|u|_p^2 = \langle u(p), u(p) \rangle_p = \text{const.}$$

New method

Technical Lemma (T. [3]). Let $\mathcal{A}$ be a unital Real C^* -algebra and $S \rightarrow M$ a graded Real $\mathcal{A}$ -linear Dirac bundle with induced $\mathcal{A}$ -linear Dirac operator $\mathcal{D}$.

► If the higher index of $\mathcal{D}$ does not vanish, there exists a family $\{u_{\epsilon}\}_{\epsilon>0}$ of almost $\mathcal{D}$ -harmonic sections, i.e.

$$\|u_{\epsilon}\|_{L^2} = 1 \quad \text{and} \quad \|\mathcal{D}^j u_{\epsilon}\|_{L^2} < \epsilon \quad \forall j > 1, \forall \epsilon > 0.$$

► If, moreover, $\|\nabla u_{\epsilon}\|_{L^2} < \epsilon$ for all $\epsilon > 0$, then there exist constants $C, r > 0$ such that

$$\|\nabla u_{\epsilon}\|_{\infty} < C\epsilon^r \quad \forall \epsilon \in (0, 1),$$

hence the family $\{u_{\epsilon}\}_{\epsilon>0}$ is **almost constant**.

Rigidity question

How rich is the space of Riemannian metrics satisfying a certain lower scalar curvature bound?

Theorem (Llarull [1]). Let $f: (M, g_M) \rightarrow (S^n, g_{\text{round}})$ be a smooth map of non-zero $\hat{A}$ -degree and $n \geq 2$. Then

$$\left. \begin{array}{l} \text{scal}_M \geq n(n-1) \\ g_M \geq f^* g_{\text{round}} \end{array} \right\} \implies \begin{array}{l} f \text{ is a Riem. submersion} \\ \text{with } \text{scal}_M = n(n-1). \end{array}$$

Goette and Semmelmann [2] generalized Llarull's rigidity theorem to a broader class of target manifolds. We establish two independent further generalizations concerning

► the **index theoretical condition** and

► the **rigidity conclusion**.

Main Theorem (Goette-Semmelmann [2], T. [3], Riedler-T. [4]). Let $f: M \rightarrow N$ be a **spin** map between two closed connected Riemannian manifolds of dimension m and n , respectively. Suppose

► $\mathcal{R}_N \geq 0, \text{Ric}_N > 0$ and

► $\chi(N) \cdot \text{deg}_{\text{hi}}(f) \neq 0 \in KO(C^*\pi)$.

Then the following implication holds:

$$\left. \begin{array}{l} \text{scal}_M \geq \text{scal}_N \circ f \\ g_M \geq f^* g_N \end{array} \right\} \implies \left\{ \begin{array}{l} M \cong N \times F \text{ (locally isometric)} \\ \text{for some Ricci-flat } F \text{ and } f \cong \text{pr}_1. \end{array} \right.$$

Definition. ► The map f is called **spin** if

$$w_i(TM) = f^*(w_i(TN)) \quad (i = 1, 2).$$

► The **higher mapping degree** of f is defined for a regular value p of the map f via

$$\text{deg}_{\text{hi}}(f) := \text{ind}(\mathcal{D}_{S^{f^{-1}(p)} \otimes \mathcal{L}(M)}|_{f^{-1}(p)}) \in KO(C^*\pi).$$

Proof. The spin map f gives rise to a graded Real $\mathbb{C}l_{m,n} \otimes C^*\pi$ -linear Dirac bundle $\mathcal{S}_{\mathcal{L}} := SM \otimes f^*SN \otimes \mathcal{L}(M)$ with Clifford multiplications

$$c: TM \rightarrow \text{End}(\mathcal{S}_{\mathcal{L}}) \quad \text{and} \quad \bar{c}: f^*TN \rightarrow \text{End}(\mathcal{S}_{\mathcal{L}}).$$

The induced Dirac operator $\mathcal{D}_{\mathcal{L}}$ satisfies

► $\text{ind}(\mathcal{D}_{\mathcal{L}}) = \text{deg}_{\text{hi}}(f) \cdot \chi(N) \neq 0$ and

► $\mathcal{D}_{\mathcal{L}}^2 \geq \nabla^* \nabla + \frac{1}{4}(\text{scal}_M - \text{scal}_N \circ f)$.

By the Technical Lemma there exists a family of **almost constant** sections $\{u_{\epsilon}\}$, and it follows that f is a Riemannian submersion with $\text{scal}_M = \text{scal}_N \circ f$. Moreover, there exist $C, r > 0$ such that

$$\|(c(\omega) - \bar{c}(f^*\bar{\omega}))u_{\epsilon}\| \leq C\|\omega\|\epsilon^r$$

for all $\epsilon \in (0, 1)$ and all $\bar{\omega} \in \text{im}(\mathcal{R}_N)$ with horizontal lift ω . It follows that the O'Neill tensors A and T of the map f vanish. $\square$

New examples. The Main Theorem applies to the maps

► $\text{pr}_1: S^{2n} \times T^k \rightarrow S^{2n}$ and $\text{pr}_1: \mathbb{R}P^{2n} \times \Sigma^k \rightarrow \mathbb{R}P^{2n}$.

References

- [1] M. Llarull. "Scalar curvature estimates for $(n+4k)$ -dimensional manifolds". In: *Differential Geometry and its Applications* 6.4 (1996).
- [2] S. Goette and U. Semmelmann. "Scalar curvature estimates for compact symmetric spaces". In: *Diff. Geom. Appl.* 16.1 (2002).
- [3] T. Tony. "Scalar curvature rigidity and the higher mapping degree". In: *Journal of Functional Analysis* 288.3 (2025).
- [4] O. Riedler and T. Tony. *Scalar-rigid submersions are Riemannian products*. Preprint, 2026. arXiv: 2601.14117.

